



SolidWorks Project Report

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Design and Modelling of a 3D Y-joint using SolidWorks

Objective:

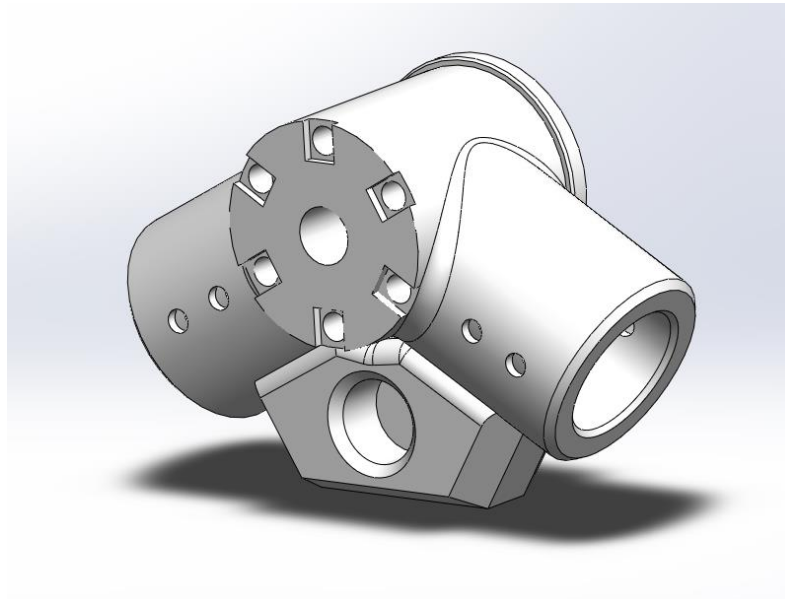
The objective of this lab is to design and model a three-dimensional Y-joint using CAD software. This experiment aims to provide an understanding of basic solid modeling techniques such as sketching, extrusion, shelling, and cutting operations, as well as to visualize a mechanical component used in piping systems.

Introduction:

In this experiment, a 3D Y-joint was designed and modeled using SolidWorks CAD software. The Y-joint is a mechanical component that is commonly used to split or combine flows in piping and duct systems. The experiment focused on applying advanced CAD techniques like surface sweeps, multi-directional extrusion, and feature-based modeling to create a practical and accurate representation of the component. Finishing methods like fillets, chamfers, and extrude cuts were also employed to improve the model's appearance, structural integrity, and manufacturability. This lab prepares students for practical mechanical design applications while assisting them in understanding the principles of 3D modeling.

Theory:

A mechanical component that divides or merges flow is called a Y-joint. It is widely used in air ducts, fluid pipelines, and mechanical assemblies. When designing such components, engineers can use CAD software to examine dimensions and shape before manufacturing.



Methodology/Modelling Approach:

The 3D Y-joint was modeled in SolidWorks using a feature-based design approach. The joint's connected branches were first created using the Surface Swept Base feature to achieve smooth geometry. Next, a circular sketch was extruded in both directions on the front plane to form the main body. Features like the mounting base and hollow structure were added using sketches, Extrude Boss/Base, and shell operations. Openings and edges were polished with fillets, chamfers, and extrude cuts. Accurately visualizing the Y-joint and practicing solid and surface modeling techniques were made possible by this methodical approach.

Procedure:

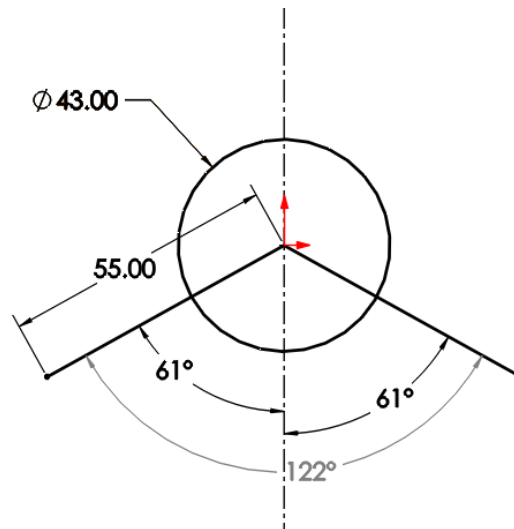
The 3D Y-Joint was designed using the following procedures:

Step 1:

After opening SolidWorks, a new Part document was made. In order to guarantee precise dimensions during modeling, appropriate units (MMGS) were chosen.

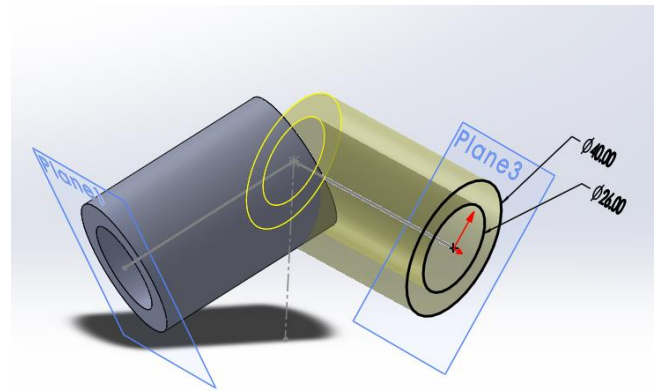
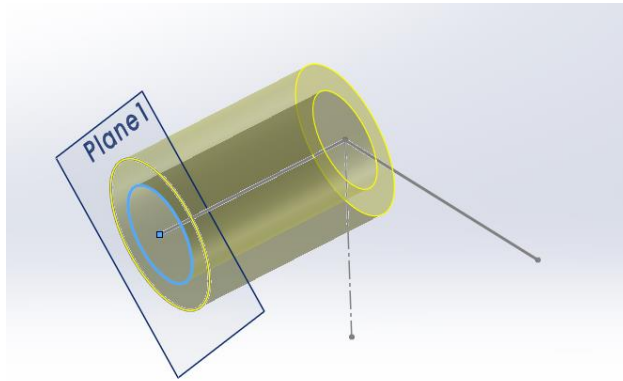
Step 2:

They were analyzed in order to prepare reference planes for surface modeling. Sketch paths were used to define the shape and direction of the Y-joint branches.



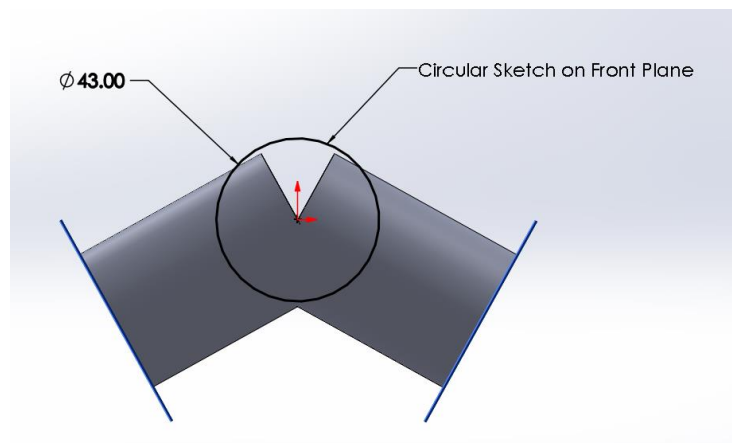
Step 3:

The Sheet Metal Base Flange/Tab feature was used to create the Y-shaped geometry with smoothly connected joint branches. Without utilizing the Surface Swept Base feature, this method permitted smooth transitions between both branches and controlled curvature.



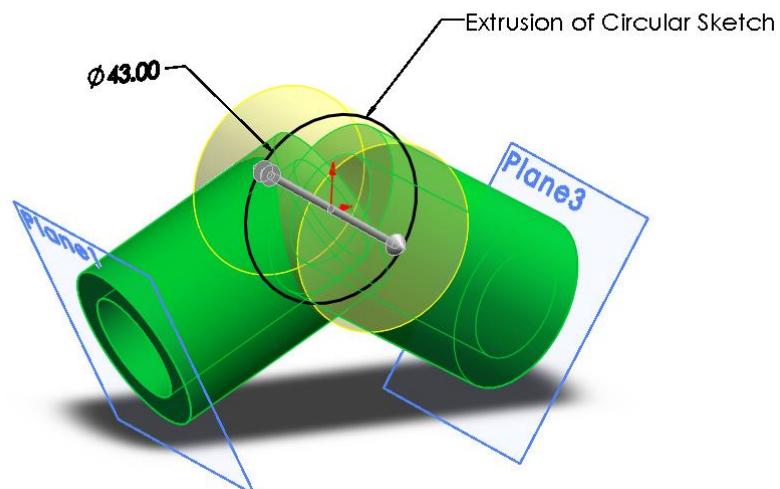
Step 4:

A circular sketch was drawn on the Front Plane to create the main pipe section of the Y-joint. The sketch was fully constrained to maintain dimensional accuracy.



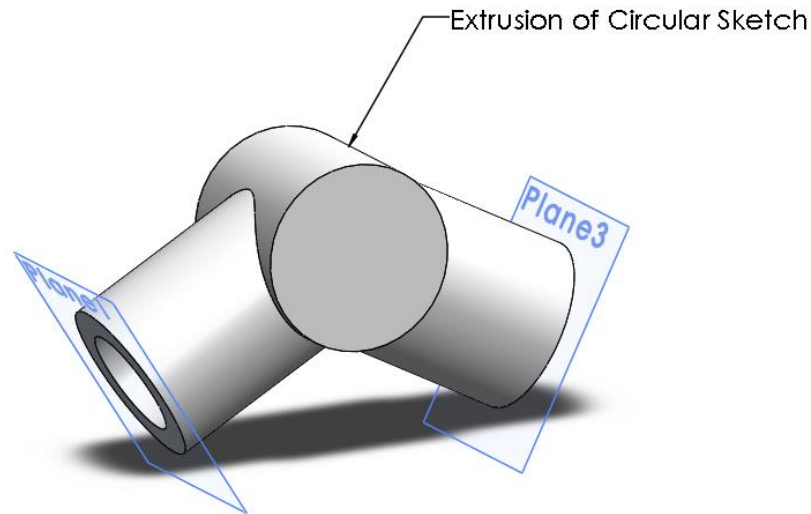
Step 5:

The circular sketch was extruded in two opposite directions using the Extrude Boss/Base feature. This created the main body of the Y-joint extending equally on both sides.

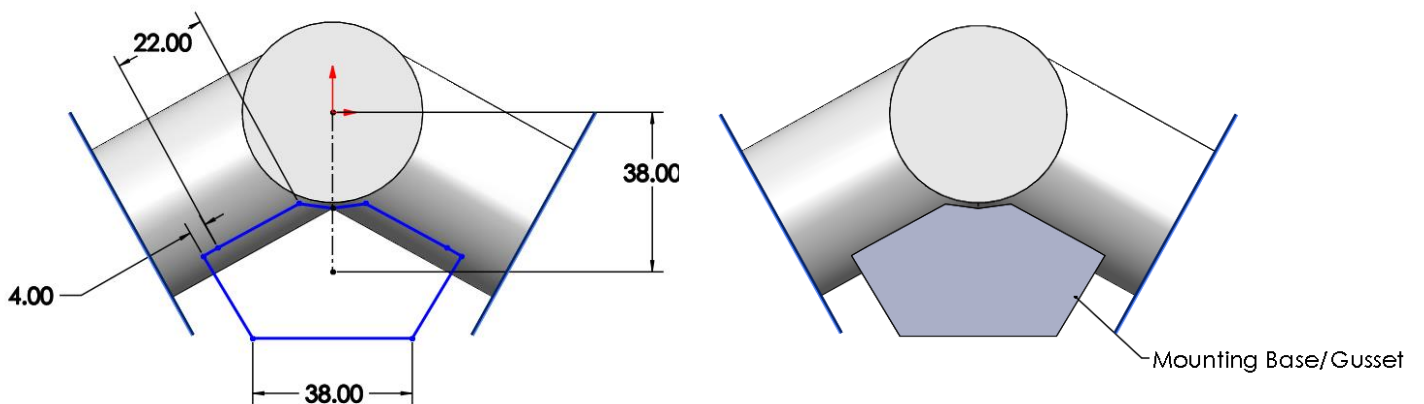


Step 6:

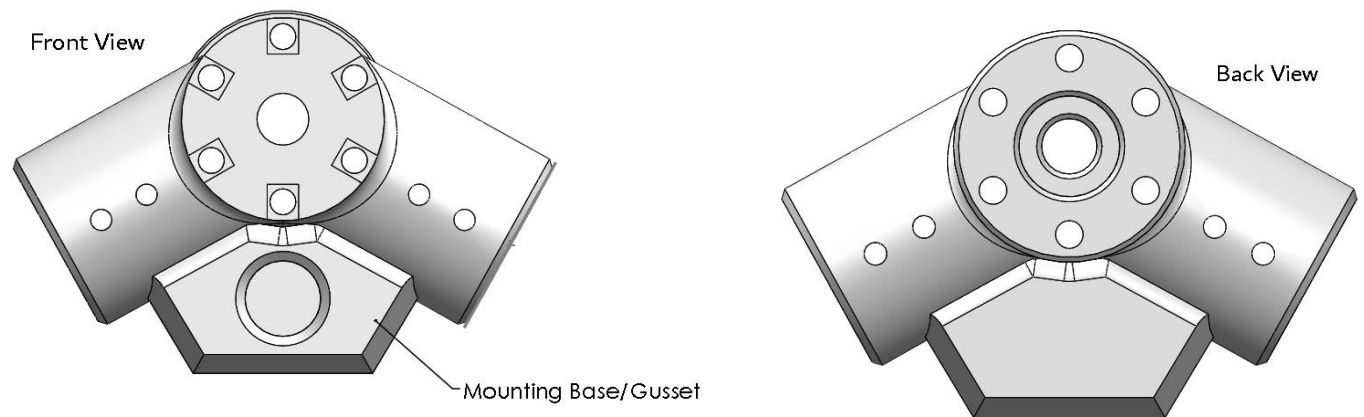
The extruded solid was merged with the previously created surface geometry to form a complete Y-joint structure.

**Step 7:**

A mounting base was added at the bottom to provide stability and structural support for the Y-joint. A drawing of the mounting base (gusset) was created and suitably constrained on the bottom reference plane. The sketched profile was then extruded in both directions using the Extrude Boss/Base feature and combined with the Y-joint main body.

**Step 8 (Result):**

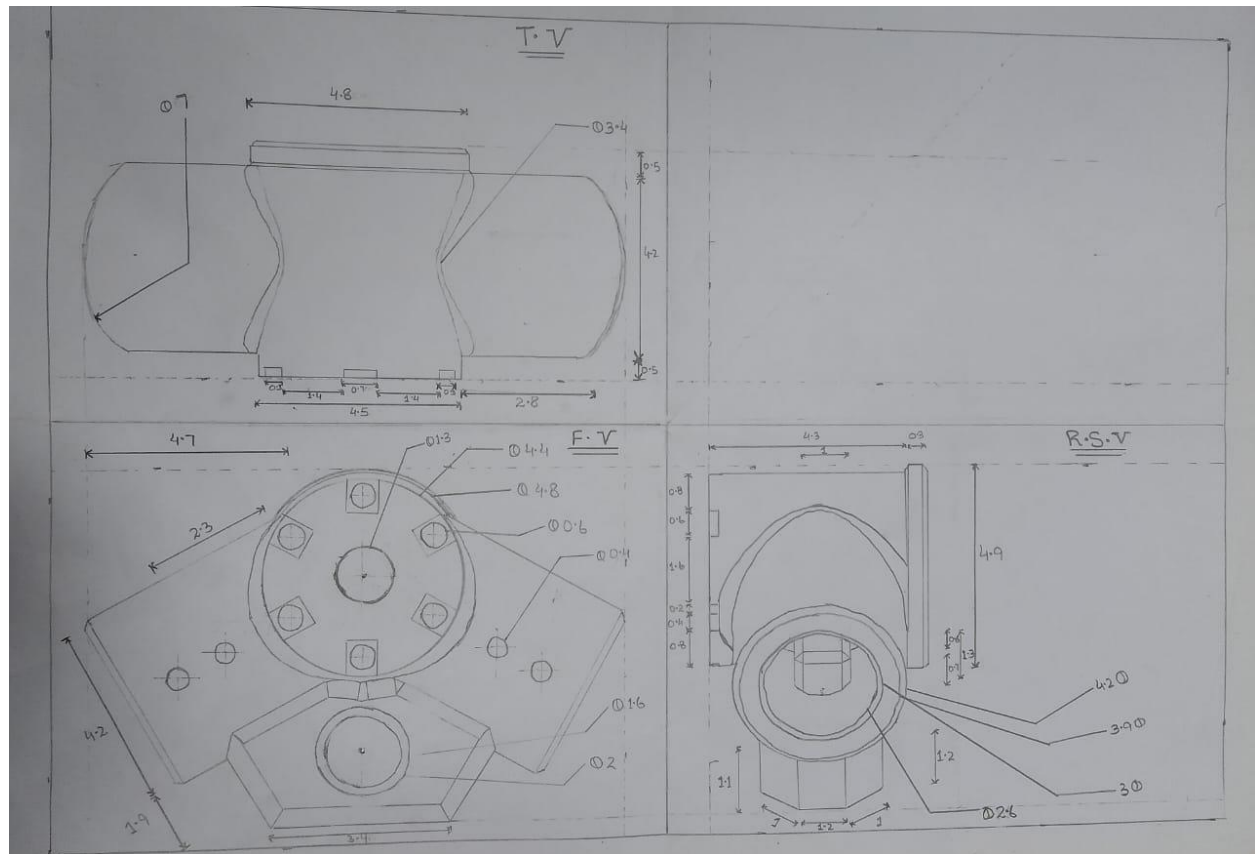
Fillets and chamfers were added to smooth edges, and extrude cuts were used to remove extra material and create the necessary openings. After a few small geometry adjustments, the final Y-joint model was reviewed and saved.



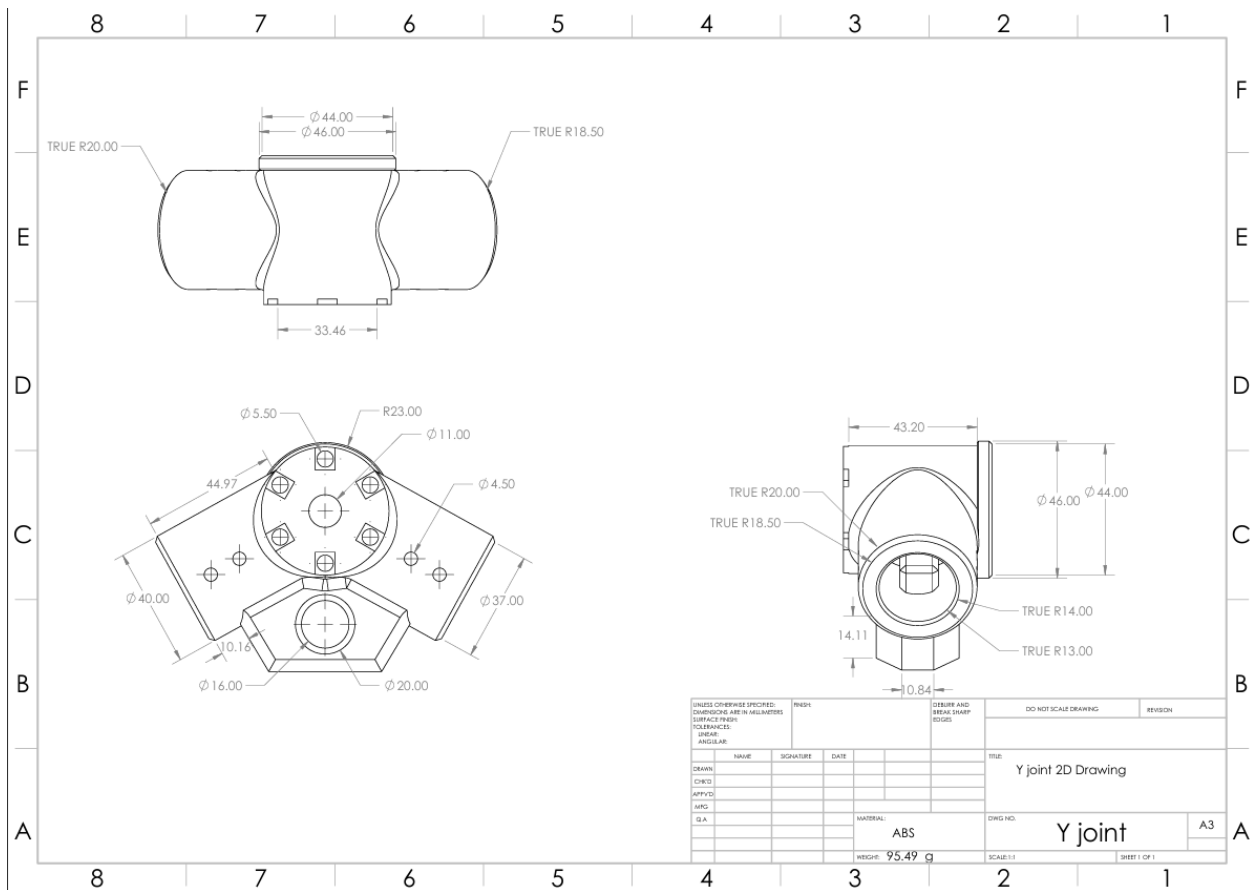
Discussion:

The 3D Y-joint model demonstrated the practical applications of SolidWorks' surface and solid modelling. It was difficult to hollow the joint uniformly and join branches smoothly. Finishing elements like fillets, chamfers, and extrude cuts improved CAD skills and feature-based design knowledge while also improving the model's quality.

2D Sketch of Y joint:



2D Drawing SolidWorks Y joint:



Conclusion:

A functional 3D Y-Joint was made. This experiment enhanced solid modeling, visualization, and the use of CAD features like extrusion, shelling, and cutting operations.

Design and Modeling of a 3D Car Dashboard Using CAD Software

Objective:

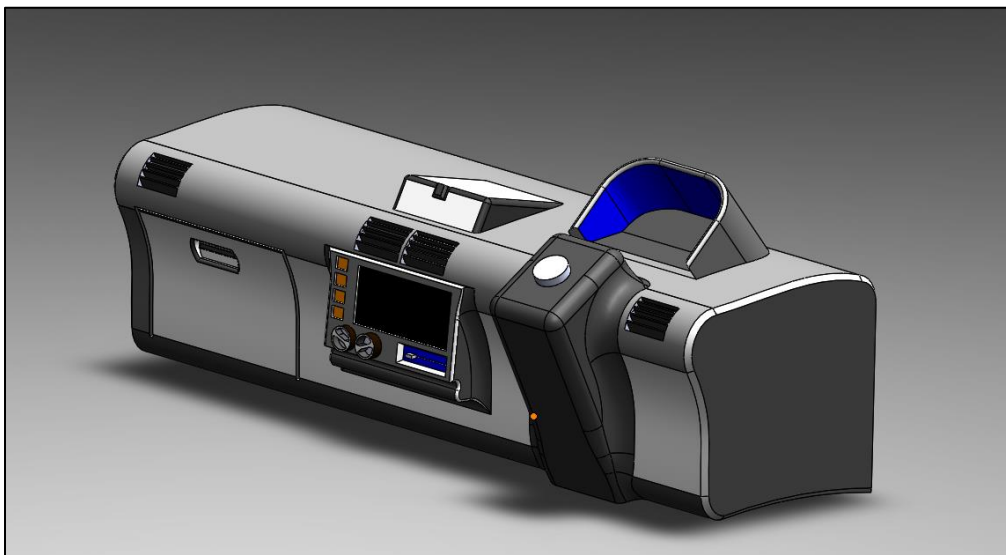
The objective of this lab is to create a realistic model with practical features by designing and modelling a 3D car dashboard using advanced CAD techniques.

Introduction:

In this project, a 3D car dashboard was designed and built using SolidWorks. There were numerous features on the car dashboard, such as instrument panels, meters, and indicators. The lab project's objective was to create a 3D dashboard model using the Computer Aided Design (CAD) modeling tools Sketch, Extrude, Solidify/Shaped, and Cut. The capacity to prepare for automotive design and work with 3D models is enhanced.

Theory:

Dashboards are automotive display/driver control devices that allow users to interact with their vehicles through graphical user interfaces. CAD provides tools for creating a three-dimensional representation of dashboards, ensuring that they are both aesthetically pleasing and easy to use. Solid modeling features like extrusion, shelling, filleting, and cutting can help you create an accurate and practical dashboard.



Methodology/Modelling Approach:

The car dashboard was modeled in SolidWorks using a feature-based design approach. A reference image served as a guide for the overall layout and dimensions. The main dashboard body was made by extrusion and sketching, and the shell feature was used to hollow it out. Important features like vents, steering wheel base, meter housing, LED screen, buttons, and dashboard box were added using sketches, extrude, and cut operations. Chamfers and fillets were used as finishing touches to improve the geometry and smooth the edges. The 3D dashboard model could be constructed logically and methodically thanks to this technique.

Procedure:

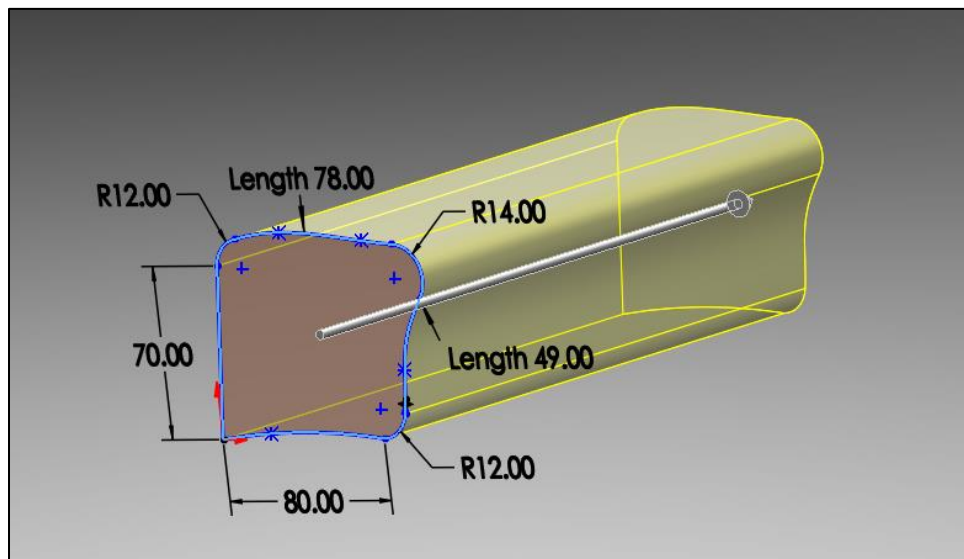
The 3D Car dashboard was designed using the following procedures:

Step 1:

After opening SolidWorks, a new Part document was made. In order to guarantee precise dimensions during modeling, appropriate units (MMGS) were chosen.

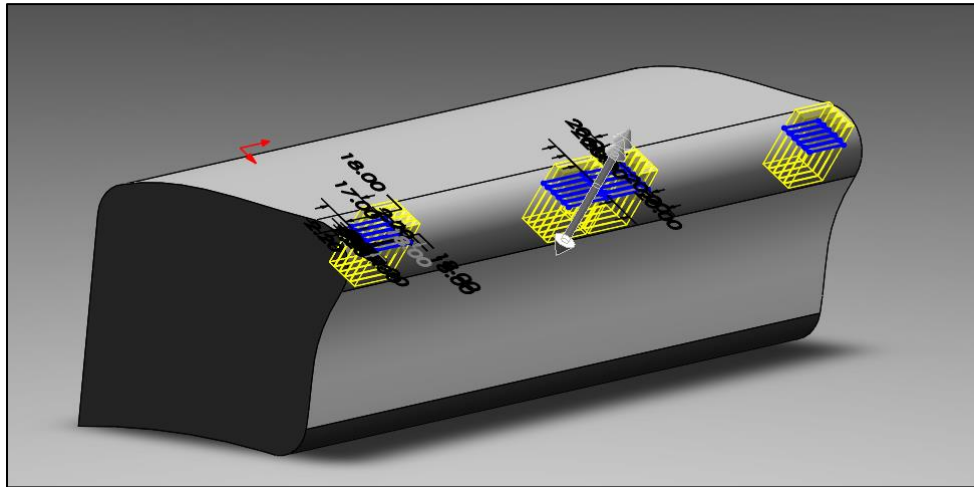
Step 2:

The main dashboard base was drawn in two dimensions on the reference plane. Using the Extrude Boss/Base feature, the sketched profile was fully constrained and extruded to a length of 270 mm to create the dashboard's main body.



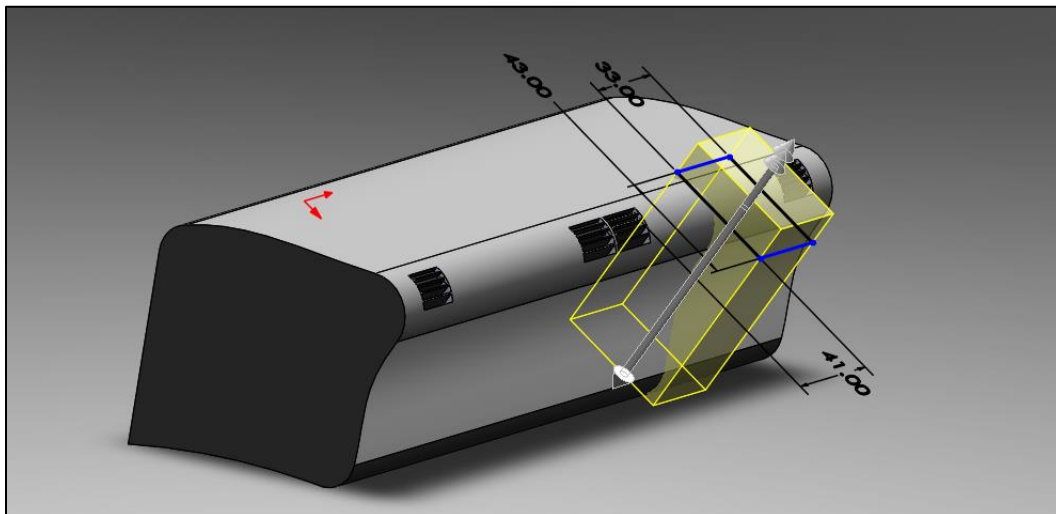
Step 3:

The Shell feature was used to hollow the dashboard body and create a uniform wall thickness. After shelling, 2D drawings of the air vents were created on the front surface, and vent openings were created using the Extrude Cut feature.



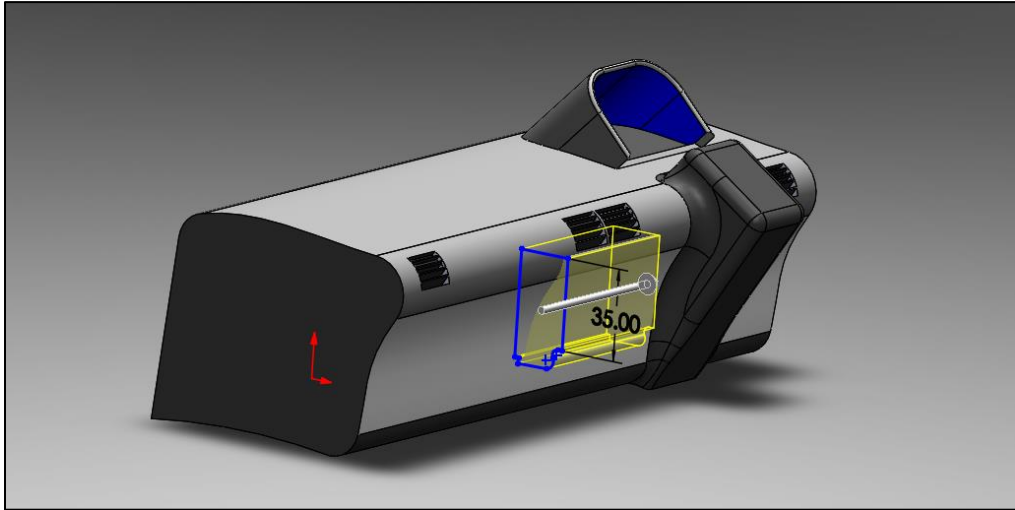
Step 4:

There is now a steering wheel base on the dashboard. The steering column base was created by extruding the supporting profile that was drawn on the appropriate plane without the steering wheel.



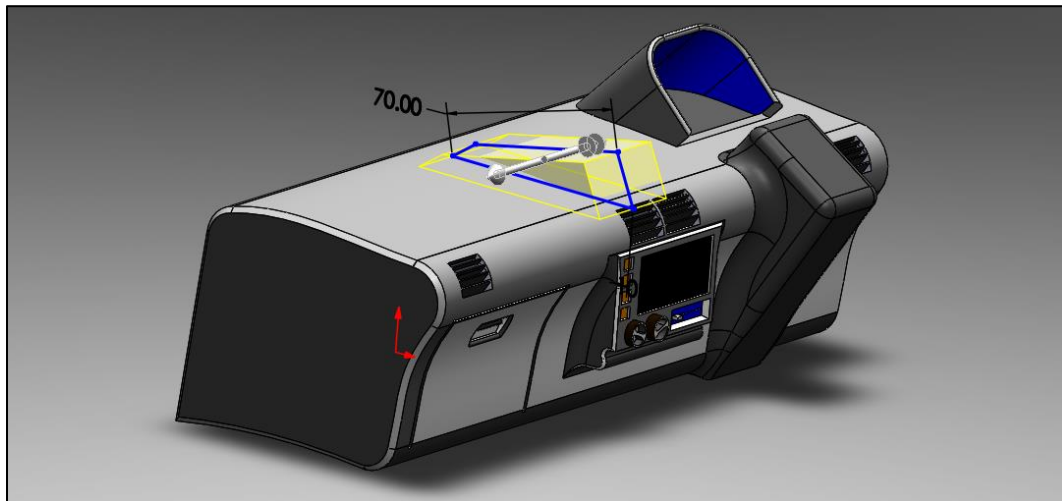
Step 5:

A central LED display area could be modeled by drawing the screen outline on the dashboard surface. The screen recess and surrounding button areas were created using Extrude Boss/Base and Extrude Cut operations.



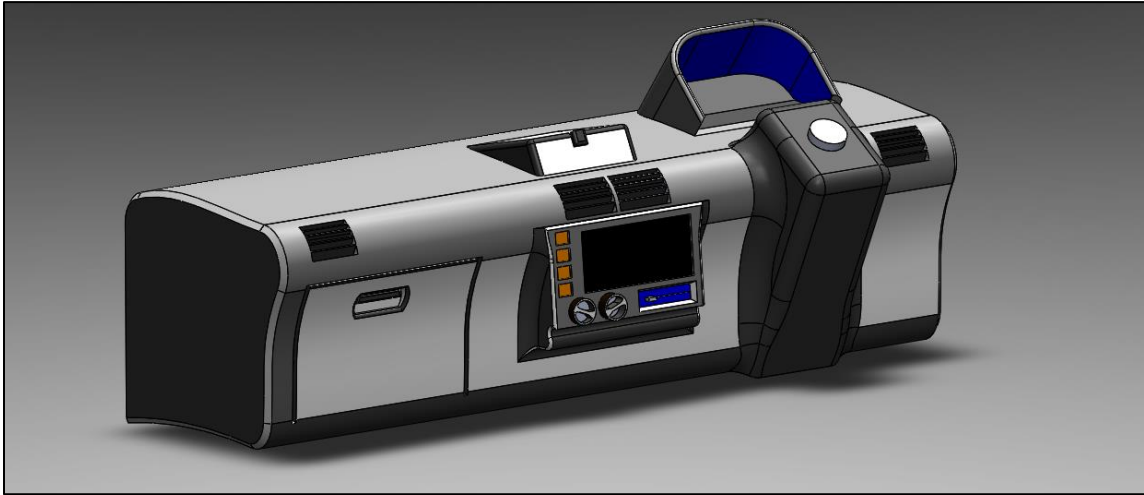
Step 6:

The lower portion of the dashboard was sketched and then extruded to create the box structure to add a dashboard storage box.



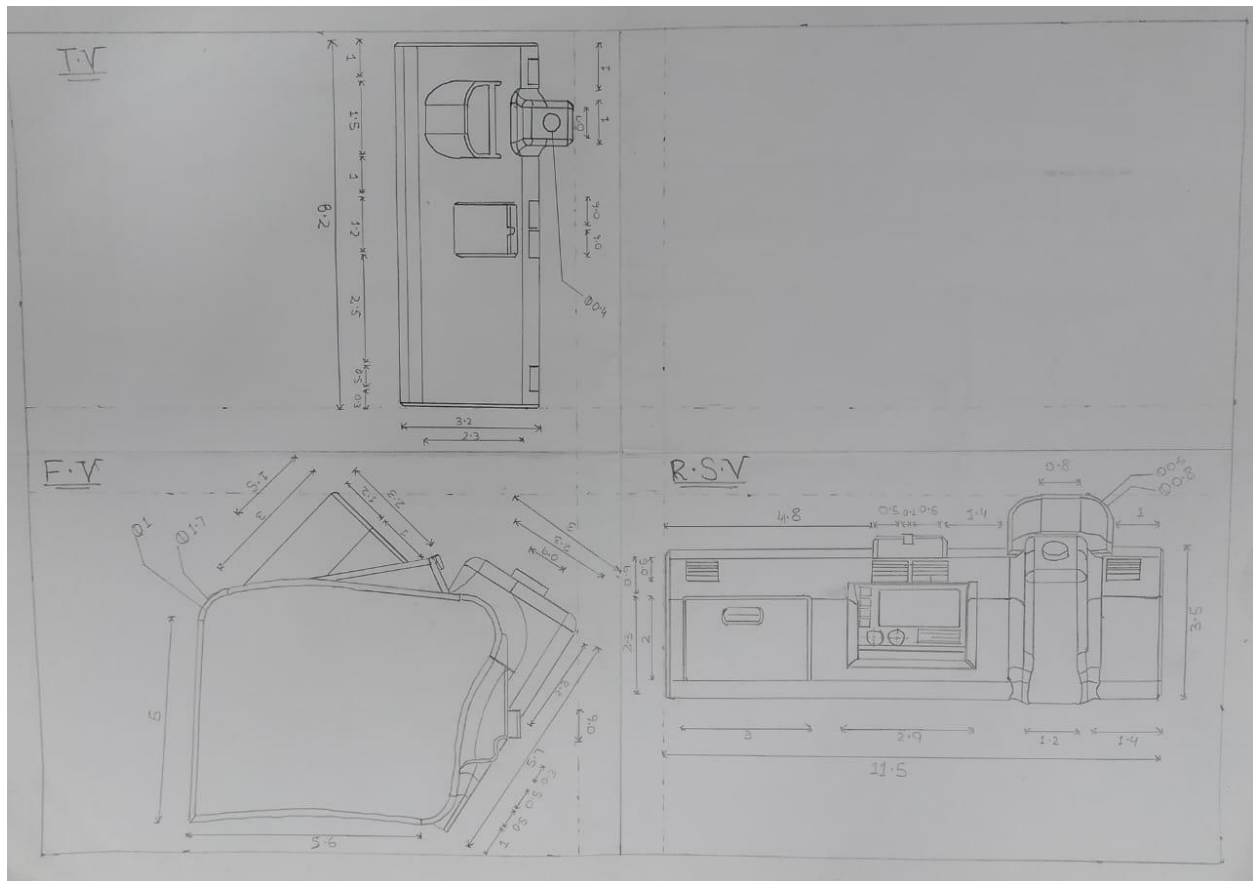
Step 7(Result):

Sharp edges were smoothed out using finishing techniques like chamfers and fillets. To improve the geometry, extra extrude cut features were used when needed. The completed dashboard model was then reviewed and saved.

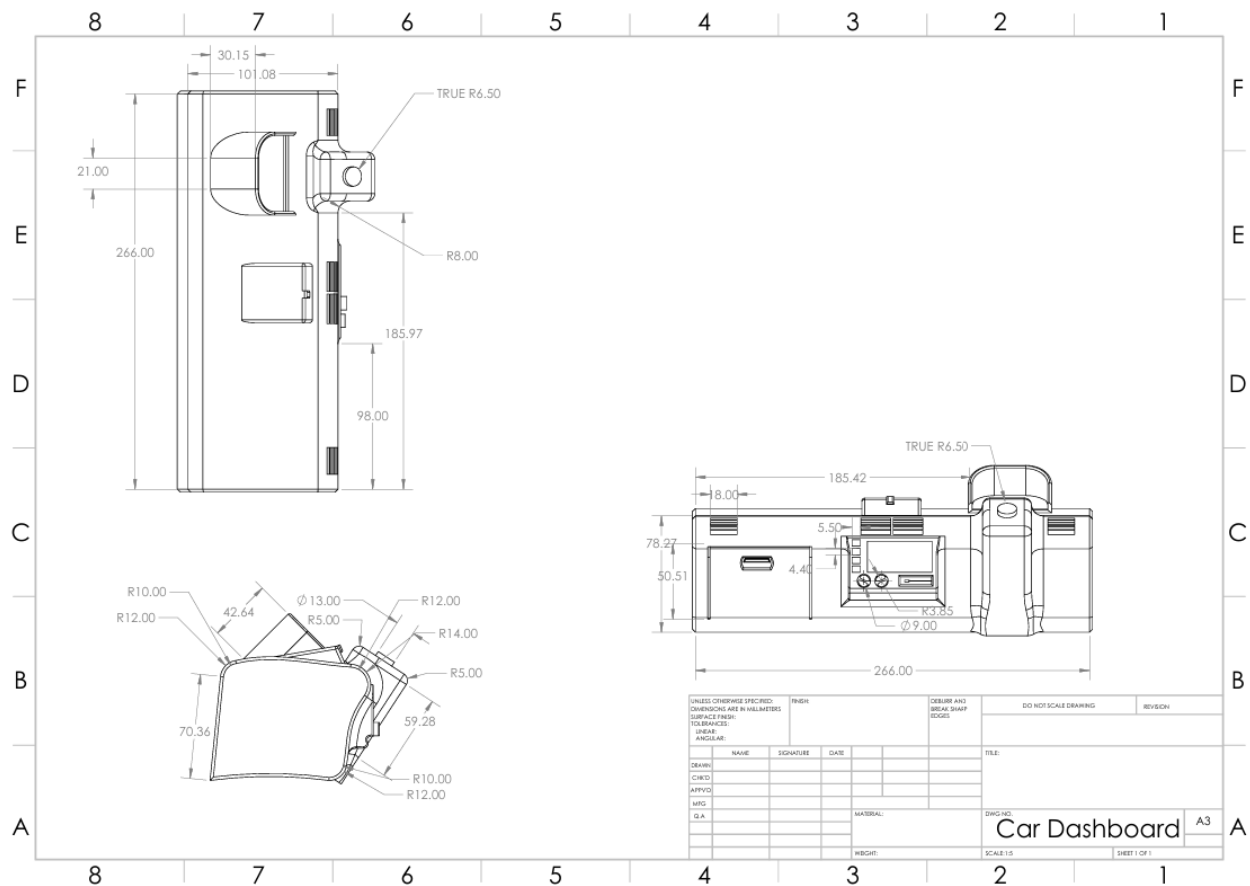
**Discussion:**

The dashboard model allowed for practice with extrusion, shelling, and cutting features. The development of vents, meter space, LED screens, and buttons improved layout planning and 3D modeling skills.

2D Sketch of Car Dashboard:



2D Drawing SolidWorks Car Dashboard:



Conclusion:

SolidWorks and an online visual reference image were used to successfully create the car dashboard model. Instead of using precise manufacturer specifications, the model's dimensions were estimated visually. Rather than attaining production-level or fully 3D-printable accuracy, the main goal of this lab was to comprehend 3D modeling techniques, feature creation, and overall form development. Participants gained experience designing intricate automotive components and enhanced their practical CAD skills through this experiment.